INCH-POUND

MIL-R-52243C(ME)
17 September 1990
SUPERSEDING
MIL-R-52243B(ME)
12 September 1979

MILITARY SPECIFICATION

RETAINER, BRIDGE PIN

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of bridge pin retainer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 65.2).

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A229	- Steel Wire, Oil-Tempered for Mechanical Springs.
A713	- Steel Wire, High-Carbon Spring, for Heat-Treated Components

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The bridge retainer shall be as shown on figure 1 and as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3 in accordance with 4.3).

3.3 Materials. Material shall be as specified herein and on figure 1. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Copies of analyses and test reports for all materials shall be available for inspection. Chemical

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composition and mechanical properties of heat-treated materials shall be verified by chemical analysis and mechanical tests (see 3.3.1).

3.3.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the bridge pin retainer may be newly fabricated from recovered materials to the maximum extent practicable, provided the bridge pin retainer produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the bridge pin retainer.

3.4 Length. The length of the bridge pin retainer shall be as shown in figure 1, as specified (see 6.2).

3.5 Identification marking. The bridge pin retainer shall be identified in accordance with MIL-STD-130.

3.6 Workmanship.

3.6.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. All bends shall be made with controlled means to insure uniformity of size and shape and shall be made to a radius which will not cause cracking at the extreme outer fibers. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either

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indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as applicable.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article bridge pin retainer shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. A lot shall be accepted if 0 defects are found and rejected if 1 or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1.

4.5 Inspection procedure.

4.5.1 Examination. The bridge pin retainer shall be examined as specified herein for the following defects:

MAJOR

101. Nonconformance to the dimensions shown on figure 1.
102. Material not as specified.
103. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the bridge retainer pin.
104. Heat treatment of bridge pin retainer not as specified.
105. Bridge pin retainer not coated as specified.
106. Workmanship not as specified.

MINOR

201. Identification marking incorrect, missing, or not as specified.

4.6 Inspection of packaging.

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4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. A lot shall be accepted if 0 defects are found and rejected if 1 or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects.

107.	Containers not as specified for level A, B, or C. Each incorrect container shall be considered one defect.
108.	Retainers of unlike size packed together for level A, B, or C.
109.	Quantities packed together exceed the limitations specified for level A, B, or C.
110.	Strapping not as specified for level A.
111.	Marking missing, illegible, incorrect, or incomplete for level A, B, or C.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The retainers of like size shall be packed together in a close fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.1.2 Level B. The retainers of like size shall be packed together as specified in 5.1.1 except the box shall be domestic type. As an alternate, the retainers of like size may be packed together in a close-fitting fiber-board box conforming to PPP-B-636, V3c, style CSSC. The gross weight or size of the fiberboard box shall not exceed the limitations of the box specification. Box closure and strapping of either the wood or fiberboard box shall be in accordance with the appendix to the applicable box specification.

5.1.3 Level C. The retainers of like size shall be packed together as specified for level B except the box shall conform to PPP-B-636, class domestic.

5.2 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The bridge pin retainers are intended for use in securing bridge pins in fixed and floating military bridges.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of the first article model (see 6.3).
d. Length of bridge pin retainer required (see 3.4).
e. Level of packing required (see 5.1).
f. Any special marking required (see 5.2).

6.3 First article. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation. Any change or deviations of production bridge pin retainers from the approved first article during production will be subject to approval by the contracting officer. Approval of the first article will not relieve the contractor of his obligations to furnish bridge pin retainers conforming to this specification.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (key word) listing.

Bridge
Fixed bridges
Floating bridges
Retainer bridge pins

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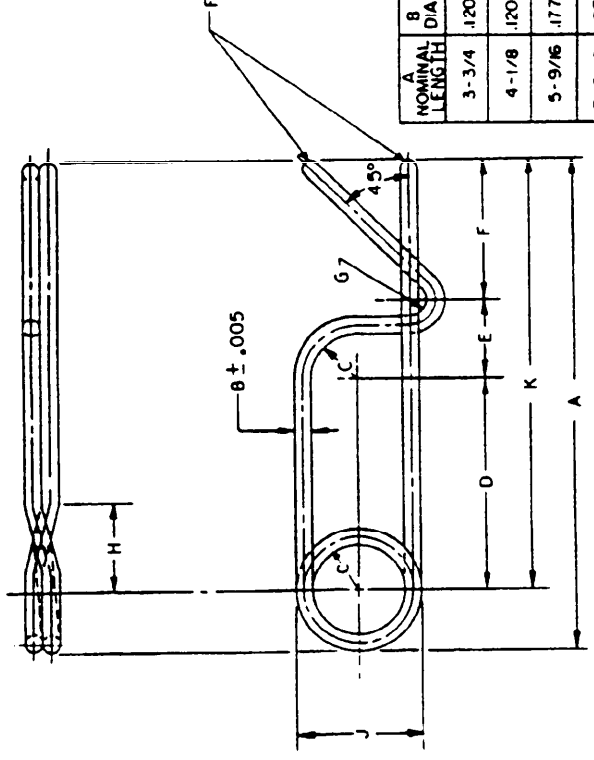
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NOTES:
1. SPRING STEEL WIRE SHALL CONFORM TO ASTM A229, CLASS I OR ASTM A713.
2. AFTER FORMING INTO SHAPE, THE RETAINER MADE OF ASTM A713 SPRING WIRE SHALL BE HEAT-TREATED TO PRODUCE A MINIMUM YIELD STRENGTH OF 190,000 PSI AND AN ULTIMATE TENSILE STRENGTH WITH THE RANGE OF 200,000 PSI (MINIMUM) TO 225,000 PSI (MAXIMUM).
3. AFTER THE HEAT TREATMENT, COAT THE RETAINER WITH A CORROSION PREVENTIVE SOLVENT CUT BACK COATING, HARD FILM TYPE, IN ACCORDANCE WITH MIL-C-16173, GRADE 1.
4. UNLESS OTHERWISE NOTED, ALL DIMENSIONS ARE IN INCHES, AND TOLERANCES ± 1/16.



A NOMINAL LENGTH	B DIA.	C RAD.	D	E	F	G RAD	H	J APPROX.	K	COMMONLY USED WITH CONN PIN OF DIA.	NSN
3-3/4	.120	5/16	1-3/4	9/16	1	1/8	3/4	7/8	3-5/16	UP TO 1	5420-060-00-7030
4-1/8	.120	3/8	2	5/8	1	1/8	3/4	1	3-5/8	1-1-1/2	5420-060-00-7031
5-9/16	.177	1/2	2-5/8	7/8	1-3/8	3/16	1	1-3/8	4-7/8	1-1/2-2	5420-061-00-2873
7-13/16	.177	3/4	3-7/8	1-1/8	1-7/8	3/16	1-1/2	1-7/8	6-7/8	2-3	5420-060-00-7032

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FIGURE 1. Retainer, bridge pin

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-52243C(ME)	900917

3. DOCUMENT TITLE

Retainer, Bridge Pin

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-5717
	(2) AUTOVON 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466

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